Chow Varieties

25110840012 上海数学中心 谢景炎

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1 Introduction

The study of parameter spaces for algebraic cycles is a foundational cornerstone of algebraic geometry. While the Hilbert scheme successfully parameterizes closed subschemes of a projective variety by keeping track of the scheme-theoretic structure, there is frequently a need to parameterize uniruled or effective algebraic cycles where the nilpotent structure is forgotten and only the formal sum of subvarieties and their multiplicities is retained. The parameter space that achieves this is called **Chow variety**.

This note following the Kollár's book [1] explores the representability of the Chow functor, tracing its classical roots to its modern functorial formulation. We will outline the construction and representability in characteristic 0 via the Chow form, and subsequently analyze the geometric and algebraic pathologies that arise in characteristic $p > 0$, highlighting why classical approaches fail and how the moduli problem must be adjusted.

1.1 Historical context

The concept of parameterizing varieties of a given dimension and degree dates back to the classical era of algebraic geometry. The foundations were laid in 1937 by W. L. Chow and B. L. van der Waerden in their seminal paper *Zur algebraischen Geometrie IX*, where they introduced the **Chow form** (or often called the Cayley form) to assign a set of homogeneous coordinates to any projective variety.

By associating an algebraic cycle to a specific point in a projective space, they demonstrated that the set of all effective cycles of a fixed dimension and degree forms a closed algebraic set—the Chow variety. In the 1950s and 60s, Grothendieck incorporated the Chow variety into the modern scheme-theoretic framework, notably establishing the **Hilbert-Chow morphism**, which provides a natural map from the Hilbert scheme of closed subschemes to the Chow variety of cycles. However, defining the Chow variety as representing a well-behaved functor of *families of cycles* over arbitrary base schemes remained a delicate problem, leading to extensive modern treatments by mathematicians such as D. Mumford, J. Kollár, and B. Rydh.

1.2 Representability in characteristic 0

In characteristic 0, the representability of the Chow variety can be established robustly using the classical theory of Chow forms and symmetric products. The main steps to construct the Chow variety $\mathrm{Chow}_{d,d'}(\mathbb{P}^n_S)$ (parameterizing cycles of dimension d and degree d' in $\mathbb{P}^n_S$) and prove its representability through following main steps:

- **Step 1: Construction of the** Ch **Map.** We begin by defining a natural transformation from the functor of families of cycles to a projective space. For a given family of cycles, we construct the Chow map (the Ch morphism), which assigns to each cycle its corresponding point in a projective space $\mathbb{P}(V)$. This projective space parameterizes effective Cartier divisors of a specific degree on the relative Grassmannian.

- **Step 2: The Chow Form.** The precise algebraic object underlying the Ch map is the Chow form. For an individual irreducible k-dimensional subvariety $X \subset \mathbb{P}^n_S$, we consider the incidence correspondence between X and the Grassmannian $\mathrm{Gr}(n-k-1, n,)$. The locus of $(n-k-1)$-planes that intersect X forms a hypersurface in the Grassmannian. The single defining section of this hypersurface is the Chow form F_X. This construction extends additively to formal cycles.

- **Step 3: Construction of the Universal Family.** To show representability, we must construct a universal family of cycles over the closed subscheme defined by the image of the Ch map. This involves constructing a closed subscheme $\mathcal{U} \subset \mathrm{Chow} \times \mathbb{P}^n_S$ that universally parameterizes the cycles, derived from the universal properties of the incidence variety and the zero-locus of the universal Chow form.

- **Step 4: Completing the Proof.** Finally, we prove that the Ch map factors through a well-defined closed subscheme (the Chow variety) and that pulling back the universal family $\mathcal{U}$ via this map recovers the original family of cycles.

2 Family of Cycles

2.1 Algebraic cycles

In this note, we assume all schemes to be noetherian. Recall some basic definitions and properties for algebraic cycles first.

Definition 2.1. *Let X be of finite type over S. A d-dimensional algebraic cycle on X/S is a formal linear combination $\sum_i a_i[V_i]$, where V_i are irreducible closed subschemes of X over a 0-dimension closed subscheme of S. Denote $Z_d(X)$ to be the set of all d-dimensional algebraic cycles on X/S. We say a cycle is effective if $a_i \geq 0$ for all i. In particular, the fundamental cycle $[X] := \sum_i m_i[X_i]$, where X_i are d-dimensional irreducible components of X and $m_i = \mathrm{length}(\mathcal{O}_{X,X_i})$.*

For a finite S-morphism $f : X \to Y$, define pushforward $f_* : Z_d(X) \to Z_d(Y)$ sending $[V]$ to $\deg(V/W)[W]$, where W is the schematic image of $f(V)$ and $\deg(V/W) = [K(V) : K(W)]$ the extension degree of function fields.

For a flat S-morphism $f : X \to Y$ of relative dimension r, define pullback $f^* : Z_d(Y) \to Z_{d+r}(X)$ sending $[W]$ to $[f^{-1}(W)]$, where $f^{-1}(W)$ is identified with $X \times_Y W$. Note that if f is surjective, then f^* is injective.

Let $Z = \sum_i a_i[V_i]$ be a d-dimensional algebraic cycle on X/S, D Cartier divisor on X. Define the degree of Z with respect to D to be $\deg_D Z = \sum_i a_i(V_i \cdot D^d)$, where $V_i \cdot D^d$ is the intersection product.

Assume X is of finite type over k. Then for any field extension K/k, $f : X_K := X \times_k K \to X$ is faithfully flat of relative dimension 0. This gives an injective pullback map $f^* : Z_d(X) \to Z_d(X_K)$. For $Z \in Z_d(X_K)$, we say Z defined over a subfield $k \subset F \subset K$ if Z lies in the image of $Z_d(X_F) \to Z_d(X_K)$.

Remark 2.1. *For convenience, in many references, we would also call formal linear combination $\sum_i a_i[V_i]$ cycle even if V_i are not of same dimension. Here we call them quasi-cycles to distinguish from cycles.*

In addition, when pull back along finite field base change, by some useful result, we can see that $f^*[V] = p^k \sum_i [V_i]$, where V_i are irreducible. In fact, p^k comes from purely inseparable part of the base change while those V_i comes from the separable part.

Lemma 2.1. *Given a fibered diagram*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & Y \end{array} \tag{1}$$

where f is finite and g is flat. Then for any $Z \in Z_d(X)$, $g^* \circ f_*(Z) = f'_* \circ g'^*(Z)$..

Proposition 2.1. *Let X be a scheme, $Z = \sum_i a_i[V_i] \in Z_d(X)$. There is a unique closed subscheme W of X such that the fundamental cycle $[W] = Z$ if one of the following conditions holds*

- *$a_i = 1$ for all i.*
- *X is nonsingular at codimension 1 and has pure dimension $d + 1$.*

Lemma 2.2. *Let X be a scheme of finite type over field k, D Cartier divisor on X. If K/k is a field extension, $X_K := X \times_k K$ with projection $p : X_K \to X$, then for all $Z \in Z_d(X)$, $\deg_{p^*D} p^*Z = \deg_D Z$.*

Proof. First note that $p^*(Z \cdot D^d) = p^*Z \cdot (p^*D)^d$ in Chow group. Suffice to show for all closed point $x \in X$ and $f^*[x] = \sum_i m_i[y_i]$, we have that $\chi(X, x) = \sum_i m_i \chi(X_K, y_i)$. In fact, this immediately comes from the equation

$$[k(x) : k] = \sum_i \operatorname{length}(\mathcal{O}_{K \times_k k(x), y_i}) \cdot [k(y_i) : K] \tag{2}$$

since $k(x) \otimes_k K \cong \prod_i \mathcal{O}_{K \times_k k(x), y_i}$. $\square$

Proposition 2.2. *Let X be a scheme of finite type over field k of characteristic p (set $p = 1$ for characteristic 0). Assume $Z = \sum_i a_i[V_i] \in Z_d(X_K)$ for some field extension K/k. Set $I_r := \{i \big| a_i = r\}$ and $\mu(r)$ the maximal integer such that $p^{\mu(r)} \mid r$. Then Z is defined over $k \subset F \subset K$ if and only if $p^{\mu(r)} \sum_{i \in I_r} [V_i]$ is defined over F for all r.*

Proof. "$\Rightarrow$ ": Assume Z is defined over F, then Z is pullback of some cycle Z'. By remark above, the coefficients after base change would be multiplied by some power of p, say p^{μ_0}. Then $\mu_0 \leq \mu(r)$ for all r and hence we get Z' is of the form $\sum_j b_j[W_j]$, where $b_j = \frac{r}{p^{\mu_0}}$ for some r.

Set $J_r = \{j \big| b_j = \frac{r}{p^{\mu_0}}\}$. Then pullback of $\sum_{j \in J_r} b_j[W_j]$ is $\sum_{i \in I_r} r[V_i]$. Hence pullback of $\sum_{j \in J_r} [W_j]$ is $p^{\mu_0} \sum_{i \in I_r} [V_i]$. As $\mu_0 \leq \mu(r)$, we get $p^{\mu(r)} \sum_{i \in I_r} [V_i]$ is defined over F for all r.

"$\Leftarrow$ ": Note that $Z = \sum_r \frac{r}{p^{\mu(r)}} \cdot (p^{\mu(r)} \sum_{i \in I_r} [V_i])$, once we know $p^{\mu(r)} \sum_{i \in I_r} [V_i]$ is defined over F for all r, we immediately get Z is also defined over F. $\square$

Corollary 2.1. *Notations as Proposition 2.2 In particular, if $p^{\mu(r)}\sum_{i\in I_r}[V_i]$ have minimal field $k(r)$, then k' the field spanned by $\cup_r k(r)$ over k is just the minimal field of Z.*

Proof. By previous proposition, it is clear that Z is defined over k'. If Z is defined over some field k'' such that $k' \not\subset k''$. Then there must be some r such that $k(r) \not\subset k''$, contradiciting to minimality of $k(r)$. Conclude that k' is minimal. □

Proposition 2.3. *Let X be a scheme over field k of characteristic p (set $p = 1$ for characteristic 0). Assume $Z = \sum_i a_i[V_i] \in Z_d(X_K)$ for some field extension K/k. Then there is a minimal field $k \subset k' \subset K$ such that Z is defined over k' if one of the following conditions holds*

- *k is of characteristic 0*
- *$\gcd(p, a_i) = 1$ for all i*
- *X is regular in codimension 1 and Z has pure codimension 1 in X.*

Proof. For first two cases, we see $\mu(r) = 0$ for all 0. Then by Proposition 2.1, we know $\sum_{i\in I_r}[V_i]$ is fundamental cycle of some closed subscheme $W_r \subseteq X_K$, which is of pure dimension without embedded points. Kollár shows that such a closed subscheme would have minimal field $k(r)$ in the section of Hilbert scheme. Clearly, minimal field of $\sum_{i\in I_r}[V_i]$ is same to the one of W_r. By Corollary 2.1, we know Z has minimal field k'.

For the last case, again by Proposition 2.1, we know Z is fundamental cycle of some closed subscheme $W \subseteq X_K$, which is of pure dimension without embedded points. Hence Z has minimal field k' same to W. □

Corollary 2.2. *Let X be a scheme over field k. Assume $Z = \sum_i a_i[V_i] \in Z_d(X_K)$ for some field extension K/k. For subgroup $G \leq \mathrm{Aut}(K/k)$, if for all $g \in G$, Z is invariant under pullback along g, then Z is defined over the field K^G.*

Proof. The assumption implies that $\mathrm{Supp}(\sum_{i\in I_r} r[V_i])$ is also invariant under pullback. Equip it with reduced scheme structure. And Kollár shows that such a closed subscheme would be defined over K^G. Denote the corresponding reduced closed subscheme by W_r.

Note that K/K^G is a Galois extension hence separable, we get $[W_r \times_{K^G} K] = \sum_{i\in I_r}[V_i]$. Thus $\sum_{i\in I_r}[V_i]$ is defined over K^G so that by Proposition 2.2, we get Z is defined over K^G. □

Definition 2.2. *Let X be a scheme of finite type over field k, K/k and L/k field extensions. We say two cycles $Z \in Z_d(X_K)$ and $Z' \in Z_d(X_L)$ are essentially equivalent if for any field F with embeddings $K \hookrightarrow F$ and $L \hookrightarrow F$, $Z_F = Z'_F$, denoted by $Z \overset{ess}{=} Z'$.*

Equivalently, $Z \overset{ess}{=} Z'$ if there is a purely inseparable extension k'/k such that $Z_{K^{p^{-\infty}}}$ and $Z'_{L^{p^{-\infty}}}$ are both defined over k' by same cycle, where the subscript $p^{-\infty}$ denote the perfect closure.

Remark 2.2. *To see the equivalence, since purely inseparable extension only changes the coefficient, it is clear that the second one implies the first one.*

Conversely, since we donot ask the embeddings $K \hookrightarrow F$ and $L \hookrightarrow F$, they may induce different $k \hookrightarrow F$. Hence for all $g \in \mathrm{Aut}(F/k)$, we see $Z_F = Z'_F$ is invariant under pullback along g. Thus by Corollary 2.2, $Z_F = Z'_F$ is defined over $F^{\mathrm{Aut}(F/k)}$ by some cycle Z'', which is a purely inseparable extension over k, denoted by k'.

Now we get $Z_F = Z'_F = Z''_F$. Set K' and L' the field spanned by k' over K and L respectively. Then since pullback along field extension is injective, we get $Z_{K'} = Z''_{K'}$ and $Z'_{L'} = Z''_{L'}$. Conclude that $Z_{K^{p^{-\infty}}}$ and $Z'_{L^{p^{-\infty}}}$ are both defined over k' by same cycle.

By definition, we immediately get the following corollaries.

Corollary 2.3. *$Z \overset{ess}{=} Z'$ if and only if $Z_{K^{p^{-\infty}}} \overset{ess}{=} Z_{L^{p^{-\infty}}}$. In particular, when characteristic 0, $Z \overset{ess}{=} Z'$ if and only if they are both defined over k by same cycle.*

Corollary 2.4. *Assume $Z \overset{ess}{=} Z'$ and both $k \hookrightarrow K$ and $k \hookrightarrow L$ factor through field extension $\widetilde{k}/k$. Then $Z \overset{ess}{=} Z'$ also over $\widetilde{k}$.*

Reason 2.1. *Since $Z_{K^{p^{-\infty}}}$ and $Z'_{L^{p^{-\infty}}}$ are both defined over k' by same cycle, they are also defined by pullback of the cycle to $\widetilde{k}^{p^{-\infty}}$.*

Corollary 2.5. *Notation as above, if $Z \overset{ess}{=} Z'$, then for any Cartier divisor D on X, we have that $\deg_D Z = \deg_D Z'$.*

Remark 2.3. *Hence two essentially same cycles have nearly same numerical property.*

Corollary 2.6. *If $Z \overset{ess}{=} Z'$, then the images of their support in X under natural projection are same.*

Reason 2.2. *For convenience, here $\mathrm{Supp}\, Z$ refers to its image in X. Since $Z \overset{ess}{=} Z'$, we get $Z_{K^{p^{-\infty}}}$ and $Z'_{L^{p^{-\infty}}}$ are both defined over some k' by cycle Z''. Then $\mathrm{Supp}\, Z = \mathrm{Supp}\, Z_{K^{p^{-\infty}}} = \mathrm{Supp}\, Z'' = \mathrm{Supp}\, Z'_{L^{p^{-\infty}}} = \mathrm{Supp}\, Z'$.*

Corollary 2.7. *Assume Z and Z' are both cycles in $Z_d(X_K)$. Write $Z = \sum_i a_i[V_i]$ and $Z' = \sum_i b_i[V_i]$. If $Z \overset{ess}{=} Z'$, then $Z = Z'$. In particular, $a_i = b_i$ for all i.*

Reason 2.3. *We may assume K is perfect, then Z and Z' are both defined over k' by some Z''. This implies that $Z = Z'$ so that $a_i = b_i$ for all i.*

2.2 Family of cycles

Lemma 2.3. *Let $T = \mathrm{Spec}\, R$ be spectrum of DVR with generic point T_g and closed point T_0. For any morphism $f : X \to T$, there is closed subscheme $F_T(X)$ of X flat over T such that for any closed subscheme $Y \subseteq X$ flat over T, we have $F_T(X) \subseteq Y$. We call $F_T(X)$ the flat closure of X over T. Moreover, the generic fiber $F_T(X) \times_T T_g$ is isomorphic to $X \times_T T_g$ as sets. So we may also call $F_T(X)$ the flat closure of generic fiber.*

Proof. Since flatness is local on the total space, we may assume $X = \mathrm{Spec}\, A$ is affine. Take the ideal $I := \{a \in A \big| \frac{a}{1} = 1 \text{ in } A_{\mathfrak{p}}, \forall \mathfrak{p} \in X \times_T T_g\}$. Claim that I is same to the R-torsion sheaf I' of A.

In fact, for all R-torsion element $a \neq 0 \in A$, assume $ra = 0$. Then $r \in \mathfrak{m}$ the maximal ideal of R and hence $r \notin \mathfrak{p}$ for all $\mathfrak{p} \in X \times_T T_g$ so that $a \in I$. Conversely, for all $a \in I$, consider the localization $A \otimes_R K$. By assumption, we know $a \in \ker(A \to A \otimes_R K)$.

Note that torsion-freeness over a DVR is equivalent to flatness, we see $A/I' \to A/I' \otimes_R K$ is injective. Thus $\overline{a} = 0$ in A/I' so that $a \in I'$. Conclude that $I = I'$ and A/I is flat over R. And for any ideal J of A such that A/J is flat over R, we get A/J is torsion-free over R so that $I \subseteq J$.

To show the generic fibers are isomorphic, consider the following short exact sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0 \tag{3}$$

Tensor by K over R, since K is flat over R, we get exact sequence

$$0 \longrightarrow I \otimes_R K = 0 \longrightarrow A \otimes_R K \longrightarrow A/I \otimes_R K \longrightarrow 0 \tag{4}$$

Thus $A \otimes_R K \cong A/I \otimes_R K$. This implies that $X \times_T T_g \cong F_T(X) \times_T T_g$, done! For general case, we can glue up I to an ideal sheaf determining our desired closed subscheme $F_T(X)$. $\square$

Definition 2.3. *Let $g : V \to W$ be a proper morphism, V irreducible and W reduced. Given a morphism $h : T \to W$ from $T = \operatorname{Spec} R$ spectrum of DVR sending T_0 to w and T_g to generic point $w_g \in W$, consider the fibered product $V_T := V \times_W T$. Take the flat closure $F_T(V_T)$ of V_T over T, then $F_T(V_T)$ is flat over T of relative dimension d. In fact, its special fiber is of pure dimension $\dim(V_T \times_T T_g)$. Set*

$$\lim_{h \to w}(V/W) := [F_T(U_T) \times_T T_0] \in Z_d(g^{-1}(w) \times_T T_0) \tag{5}$$

called the cycle-theoretic fiber of g at w at h.

Remark 2.4. *First note that by generic flatness, for general $w \in W$, $h : T \to W$ would factors through W_0 such that $g^{-1}(W_0) \to W_0$ flat. Hence $F_T(V_T)$ is just V_T itself. Thus the definition of cycle-theoretic fiber coincide with common fiber in an open dense subset of W.*

To see the special fiber is of pure dimension d, take irreducible component V' of $F_T(V_T)$. By Going-down Theorem, generic point of V' would map to T_g. While g is proper, we get V' dominates T. Then by Stacks Project 0B2J, we see the special fiber is of pure dimension d.

For convenience, in the following context, for reduced W, we would always denote W_0 to be an open dense subset (usually the maximal one) of W given by applying generic flatness to some morphisms.

Definition 2.4. *Let $g : V \to W$ be a proper morphism between reduced schemes. Assume $C(V) = \sum_i m_i[V_i]$ is quasi-cycle over V, where V_i varies over all irreducible components of V with all $m_i \neq 0$. Denote $g_i = g\big|_{V_i}$. We say $(g : C(V) \to W)$ is a quasi-well defined family of d-dimensional algebraic cycles over W if*

- *V_i maps onto some irreducible component of W for all i.*
- *Every fiber of g_i is either empty or of dimension d.*

And if moreover it satisfies

- *(Condition $\star$) For all $w \in W$, there is a cycle $g^{[-1]}(w) \in Z_d(g^{-1}(w) \times_{k(w)} k(w)^{p^{-\infty}})$ such that*
$$g^{[-1]}(w) \stackrel{ess}{=} \sum_i m_i \lim_{h \to w} (V_i/W) \tag{6}$$
for all $h : T \to W$ from $T = \operatorname{Spec} R$ spectrum of DVR sending T_0 to w and T_g to some generic point of W.

then we call $(g : C(V) \to W)$ a well defined family of d-dimensional algebraic cycles over W.

Remark 2.5. *The condition $\star$ seems rather complicated. But we will see that if W is normal and $(g : C(V) \to W)$ quasi-well defined, then condition $\star$ holds automatically. Also, when characteristic 0, we have $g^{[-1]}(w) \in Z_d(g^{-1}(w))$, which is a more natural condition.*

Corollary 2.8. *Let $(g : C(V) \to W)$ be a well defined family of cycles. Assume $h : T \to W$ is a morphism from spectrum of DVR sending T_0 to w and T_g to some generic point. Then*
$$\operatorname{Supp} g^{[-1]}(w) = \cup_i \operatorname{Supp}(\lim_{h \to w} (V_i/W)) \text{ as sets} \tag{7}$$
where all these supports refer to their image under projection to $g^{-1}(w)$ and hence the equation holds as subspaces of $g^{-1}(w)$.

Reason 2.4. *By definition of cycle-theoretic fiber, Corollary 2.6 shows that the equality holds.*

Lemma 2.4. *For $w \in W_0$ and $h : T \to W$ sending both T_0 to w and T_g to some generic point, we have that*
$$\sum_i m_i [g_i^{-1}(w)] \stackrel{ess}{=} \sum_i m_i \lim_{h \to w} (V_i/W) \tag{8}$$
so that $g^{[-1]}(w)$ is defined over $k(w)$ by $\sum_i m_i [g_i^{-1}(w)]$.

Proof. Note that $V_i \times_W T$ is flat over T, take its special fiber $V_i \times_W T_0 = g_i^{-1}(w) \times_w T_0$. Then $\lim_{h \to w} (V_i/W) = [g_i^{-1}(w) \times_w T_0]$, which is just pullback of $[g_i^{-1}(w)]$ along $T_0 \to w$. Thus
$$\sum_i m_i [g_i^{-1}(w)] \stackrel{ess}{=} \sum_i m_i \lim_{h \to w} (V_i/W) \tag{9}$$
and hence $g^{[-1]}(w) \stackrel{ess}{=} \sum_i m_i [g_i^{-1}(w)]$. By definition of essentially same, we know $g^{[-1]}(w)$ is just pullback of $\sum_i m_i [g_i^{-1}(w)]$ along $k(w)^{p^{-\infty}} \to k(w)$. $\square$

Definition 2.5. *Let X be a scheme over S, W reduced scheme over S. A well defined family of d-dimensional algebraic cycles of X/S over W/S is a well defined family of d-dimensional algebraic cycles $(g : C(V) \to W)$ such that $V \subseteq X \times_S W$ is a closed subscheme and $g : V \to W$ is the natural projection.*

Proposition 2.4. *Let X be a scheme over S, D Cartier divisor on X. Assume $(g : C(V) \to W)$ be a well defined family of d-dimensional cycles of X/S over W/S and W is connected. Then $\deg_D g^{[-1]}(w)$ is constant.*

Proof. Since W is connected, suffices to show that for any w specialization of generic point w_g, we have that $\deg_D g^{[-1]}(w) = \deg_D g^{[-1]}(w_g)$. By Corollary 2.5, only need to show

$$\deg_D(\sum_i m_i \lim_{h\to w}(V_i/W)) = \deg_D(\sum_i m_i[g_i^{-1}(w_g)]) \tag{10}$$

By Lemma 2.3, we see that $g_i^{w_g} \times_{w_g} T_g \cong F_T(V_i \times_W T) \times_T T_g$. As $F_T(V_i \times_W T)$ is flat over T, the two fibers would same degree with respect to D so that we have

$$\begin{aligned}\deg_D[g_i^{-1}(w_g)] &= \deg_D[g_i^{-1}(w_g) \times_{w_g} T_g] \\ &= \deg_D[F_T(V_i \times_W T) \times_T T_g] \\ &= \deg_D[F_T(V_i \times_W T) \times_T T_0] \\ &= \deg_D \lim_{h\to w}(V_i/W)\end{aligned} \tag{11}$$

Summing up, we are done! □

With inspiration from this result, we give a definition of degree of a well defined family of cycles.

Definition 2.6. *Let X be a scheme projective over S, $\mathcal{O}_X(1)$ given by embedding into projective bundle. For $(g : C(V) \to W)$ be a well defined family of d-dimensional cycles of X/S over W/S, if $\deg_{\mathcal{O}_X(1)} g^{[-1]}(w)$ is constant, then we call the constant d' the degree of $(g : C(U) \to W)$.*

Remark 2.6. *Hence by Proposition 2.4, when W is connected, well defined family $(g : C(V) \to W)$ has degree.*

Proposition 2.5. *Let $(g : C(V) \to W)$ be quasi-well defined family. Then $(g : C(V) \to W)$ is well defined family if and only if there are commutative diagrams*

$$\begin{array}{ccc} V'_j & \longrightarrow & \widetilde{V} \\ \downarrow{\scriptstyle \widetilde{g}_j} & & \downarrow{\scriptstyle \widetilde{g}} \\ W' & \xrightarrow{p} & W \end{array} \tag{12}$$

where $\widetilde{V}$ is a scheme proper over W such that g factors through $\widetilde{g}$ by a closed immersion $V \hookrightarrow \widetilde{V}$, $p : W' \to W$ is a proper, surjective morphism and $V'_j \subset \widetilde{V} \times_W W'$ closed subschemes flat over W' for all j, and there are integers m'_j such that

- *for all w_g generic and $w' \in p^{-1}(w_g)$, we have*

$$\sum_j m'_j[\widetilde{g}_j^{-1}(w')] \overset{ess}{=} \sum_i m_i[g_i^{-1}(w_g)] \tag{13}$$

- *for all $w \in W$ and $w'_1, w'_2 \in p^{-1}(w)$, we have*

$$\sum_j m'_j[\widetilde{g}_j^{-1}(w'_1)] \overset{ess}{=} \sum_j m'_j[\widetilde{g}_j^{-1}(w'_2)] \tag{14}$$

In particular, for all $w \in W$ and $w' \in p^{-1}(w)$, cycle over w' computes $g^{[-1]}(w)$ i.e.

$$\sum_j m'_j[\widetilde{g}_j^{-1}(w')] \overset{ess}{=} g^{[-1]}(w) \tag{15}$$

We call such a family of diagrams is associated to well defined family $(g : C(V) \to W)$.

Proof. "$\Rightarrow$ ": Assume $(g : C(V) \to W)$ is a well defined family. As $V_i \times_W W_0$ is a closed subscheme of $V \times_W W_0$ flat over W_0, it correspondends to a morphism $s_i : W_0 \to \mathrm{Hilb}(V/W)$. Consider the product of Hilbert schemes, we get a morphism

$$s : W_0 \longrightarrow \mathrm{Hilb}(V/W) \times_W \cdots \times_W \mathrm{Hilb}(V/W) \tag{16}$$

such that the ith projection is s_i. Take the schematic closure of the image $s(W_0)$, denoted by W'. Then W' projects to W, denoted by f, and $W_0 \hookrightarrow W$ would factor through $f : W' \to W$. Note that $f : W' \to W$ is proper morphism and W_0 is dense in W, f is surjective.

Take $\widetilde{V} = V$. For each projection $W' \to \mathrm{Hilb}(V/W)$, it corresponds to a closed subscheme $V'_i \subseteq V \times_W W'$ flat over W'. We then get commutative diagrams

$$\begin{tikzcd} V'_i \arrow[r] \arrow[d, "\widetilde{g}_i"] & \widetilde{V} \arrow[d, "\widetilde{g}"] \\ W' \arrow[r, "p"] & W \end{tikzcd} \tag{17}$$

Take $m'_i = m_i$. Remains to check the two essentially same conditions. By construction, we have $V_i \cong V'_i \times_{W'} W_0$. By some useful result, we first note that $f^{-1}(W_0) \to W_0$ is an isomorphism. Hence in fact, the first essentially same conditions automatically holds for all $w \in W_0$ and $w' \in f^{-1}(w)$.

For any morphism $h : T \to W$ sending T_0 to w and T_g to generic w_g, and a morphism $h' : T' \to T$ extending h, sending T'_0 to some $w' \in f^{-1}(w)$ and T'_g to $w'_g = f^{-1}(w_g)$, consider $V'_i \times_{W'} T'$ and $F_T(V_i \times_W T) \times_T T'$ as closed subschemes of $V \times_W T'$ and flat over T'. Set Z_k all the $(d+1)$-dimensional irreducible subsets of $V \times_W T'$ dominating T'. By Going-down Theorem for flatness, we can write

$$\sum_i m_i[V'_i \times_{W'} T'] = \sum_k a_k[Z_k] \quad \sum_i m_i[F_T(V_i \times_W T) \times_T T'] = \sum_k b_k[Z_k] \tag{18}$$

for integers a_k and b_k. Base change to T'_g, note that

$$\begin{aligned} \sum_i m_i[V'_i \times_{W'} T'_g] &\overset{ess}{=} \sum_i m_i[\widetilde{g}_i^{-1}(w'_g)] \\ &\overset{ess}{=} \sum_i m_i[g_i^{-1}(w_g)] \\ &\overset{ess}{=} \sum_i m_i[g_i^{-1}(w_g) \times_{w_g} T'_g] \\ &= \sum_i m_i[F_T(V_i \times_W T) \times_T T'_g] \end{aligned} \tag{19}$$

Hence $\sum_k a_k[Z_k \times_{T'} T'_g] \overset{ess}{=} \sum_k b_k[Z_k \times_{T'} T'_g]$ over $k(w_g)$. By Corollary 2.7, this implies that $a_k = b_k$. Thus

$$(g^{[-1]}(w))_{T'_0} = \sum_i m_i[F_T(V_i \times_W T) \times_T T'_0] = \sum_i m_i[V'_i \times_{W'} T'_0] = \sum_i m_i[\widetilde{g}_i^{-1}(w') \times_{w'} T'_0] \tag{20}$$

And here in our construction, for all $w' \in f^{-1}(w)$, by some useful result, we can first find $h' : T \to W'$ sending T_0 to w' and T_g into generic fibers of f. Take $h := f \circ h$, then by argument above we get the second essentially same condition.

"$\Leftarrow$ ": For any morphism $h : T \to W$ sending T_0 to w and T_g to generic w_g, we take $w'_g \in f^{-1}(w_g)$ and L spanned by $k(w'_g)$ over $k(T_g)$. Also, we can extend discrete valuation of R to L with valuation ring R'. Set $T' = \operatorname{Spec} R'$, there is a commutative diagram

$$\begin{tikzcd} k(T_g) \arrow[r] & T \arrow[r, "h"] & W \\ L \arrow[u] \arrow[r] \arrow[rr, bend right] & T' \arrow[u] \arrow[r, dashed, "h'"] & W' \arrow[u, "f"'] \end{tikzcd} \tag{21}$$

Since f is proper, by valuative criteria, h uniquely extends to a morphism $h' : T' \to W'$ sending T'_0 to $w' \in f^{-1}(w)$. Thus since the first essentially same condition holds, by similar argument as above, we get

$$\begin{aligned} \sum_i m_i \lim_{h \to w} (V_i/W) \times_{T_0} T'_0 &= \sum_i m_i [F_T(V_i \times_W T) \times_T T'_0] \\ &= \sum_j m_j [V'_j \times_{W'} T'_0] \\ &= \sum_j m_j [\widetilde{g}_j^{-1}(w') \times_{w'} T'_0] \end{aligned} \tag{22}$$

Thus by the second essentially same condition, we get there is a cycle $g^{[-1]}(w) \in Z_d(g^{-1}(w) \times_w k(w)^{p^{-\infty}})$ such that

$$g^{[-1]}(w) \stackrel{ess}{=} \sum_i m_i \lim_{h \to w} (V_i/W) \tag{23}$$

Conclude that $(g : C(V) \to W)$ is a well defined family. In particular, since $k(w)^{p^{-\infty}}$ is purely inseparable over $k(w)$, $k(w)^{p^{-\infty}} \times_w w'$ is a subfield of $k(w')^{p^{-\infty}}$. Hence we get $\sum_j m'_j [\widetilde{g}_j^{-1}(w')] \stackrel{ess}{=} g^{[-1]}(w)$ over $k(w)$. $\square$

Corollary 2.9. *Let $\widetilde{g} : \widetilde{V} \to W$ be a proper morphism, flat of relative dimension d. Assume W is reduced and $V = \widetilde{V}^{\mathrm{red}}$. Take $C(V)$ to be fundamental quasi-cycle of $\widetilde{V}$. Then $(g : C(V) \to W)$ is a well defined family of cycles.*

Proof. First check that $(g : C(V) \to W)$ is a quasi-well defined family. Note that reduction do not change topology, since $\widetilde{g}$ is flat, generic points of V map to generic points of W. Also for any generic point $w_g \in W$, since $\widetilde{g}$ is flat of relative dimension d, $g^{-1}(w_g)$ is of pure dimension d. Note that $g_i^{-1}(w_g)$ is irreducible component of $g^{-1}(W_g)$, it is of dimension d. Hence by upper semicontinuity, any fiber dimension of g_i is at least d, which also has upper bound d.

To apply Proposition 2.5, we set $W' = W$. Take the index set $\{j\}$ to be one-point set with $V' = \widetilde{V}$ and $m = 1$. Remains to show the two essentially same conditions. In fact, there is nothing to prove for the second one.

For all generic points w, suffices to show for any generic point v_g of $g^{-1}(w)$, the multiplicities at it in both sides are equal. So we only need to consider the problem locally at w

and v_g. By taking affine neighbourhood W' of w contained in W_0 and affine neighbourhood of v_g in $g^{-1}(W')$, we may assume $W_0 = \operatorname{Spec} A$ and $g^{-1}(W_0) = \operatorname{Spec} B$ are both affine. Then w and v_g correspondend to prime ideals $\mathfrak{p}$ and $\mathfrak{q}$ respectively.

By assumption, $\mathfrak{q}$ is a minimal prime ideal of B and $B/\mathfrak{q}$ is flat over A. What we want to prove is the following equation of length

$$\operatorname{length}(B \otimes_A k(\mathfrak{p}))_{\mathfrak{q}} = \operatorname{length} B_{\mathfrak{q}} \cdot \operatorname{length}(B/\mathfrak{q} \otimes_A k(\mathfrak{p}))_{\mathfrak{q}} \tag{24}$$

As B is noetherian, by some useful result, there is a sequence of B-modules

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = B \tag{25}$$

such that for each j, there is an exact sequence

$$0 \longrightarrow M_{j-1} \longrightarrow M_j \longrightarrow B/I_j \longrightarrow 0 \tag{26}$$

for some prime ideal I_j of B. In particular, by localizing at $\mathfrak{q}$, we would see that $\mathfrak{q}$ appears in $\{I_j\}$ precisely $\operatorname{length} B_{\mathfrak{q}}$ times. Now as $B/\mathfrak{q}$ is flat over A, we have $\operatorname{Tor}_A(k(\mathfrak{p}), B/\mathfrak{q}) = 0$. Then for those j such that $I_j = \mathfrak{q}$, we get exact sequences

$$0 = \operatorname{Tor}_A(k(\mathfrak{p}), B/\mathfrak{q}) \longrightarrow M_{j-1} \otimes_A k(\mathfrak{p}) \longrightarrow M_j \otimes_A k(\mathfrak{p}) \longrightarrow B/\mathfrak{q} \otimes_A k(\mathfrak{p}) \longrightarrow 0 \tag{27}$$

Localize at $\mathfrak{q}$, we get exact sequence

$$0 \longrightarrow B_{\mathfrak{q}} \otimes_B M_{j-1} \otimes_A k(\mathfrak{p}) \longrightarrow B_{\mathfrak{q}} \otimes_B M_j \otimes_A k(\mathfrak{p}) \longrightarrow B_{\mathfrak{q}} \otimes_B B/\mathfrak{q} \otimes_A k(\mathfrak{p}) \longrightarrow 0 \tag{28}$$

Computing their length as $(B \otimes_A k(\mathfrak{p}))_{\mathfrak{P}}$-modules and summing up over j such that $I_j = \mathfrak{q}$, we get the following inequality

$$\operatorname{length}(B \otimes_A k(\mathfrak{p}))_{\mathfrak{q}} \geq \operatorname{length} B_{\mathfrak{q}} \cdot \operatorname{length}(B/\mathfrak{q} \otimes_A k(\mathfrak{p}))_{\mathfrak{q}} \tag{29}$$

To show the equality holds, we need to consider the contribution of those j such that $I_j \neq \mathfrak{q}$. Since $\mathfrak{q}$ is minimal, for all $I_j \neq \mathfrak{q}$, they would be killed by the localization at $\mathfrak{q}$, done! □

Lemma 2.5. *Assume there is a family of diagrams associated to two well defined families $(g : C(V) \to W)$ and $(g' : C(V') \to W')$. Then the two well defined family are same.*

Proof. Denote the diagrams by

$$\begin{tikzcd} \widetilde{V}_j \arrow[r] \arrow[d, "\widetilde{g}_j"] & \widetilde{V} \arrow[d, "\widetilde{g}"] \\ \widetilde{W}' \arrow[r, "p"] & \widetilde{W} \end{tikzcd} \tag{30}$$

Firstly we must have $W = \widetilde{W} = W'$. For all $w \in W$ and $w' \in p^{-1}(w)$, by Proposition 2.5, we know cycle over w' computes $g^{[-1]}(w)$ and $g'^{[-1]}(w)$. By Corollary 2.7, we see $g^{[-1]}(w) = g'^{[-1]}(w)$.

For all generic point v_g of V mapping to w_g, since $v_g \in \operatorname{Supp} g^{[-1]}(w_g) = \operatorname{Supp} g'^{[-1]}(w_g)$, we get $v_g \in V'$ and hence $V \subseteq V'$. Similarly, we have $V \supseteq V'$ so that $V = V'$ as sets. Moreover, since they are both equipped with reduced scheme structure, we get $V = V'$ as schemes. In addition, since multiplicity of $C(V)$ at v_g equal to multiplicity of $g^{[-1]}(w_g)$ at v_g, we conclude $(g : C(V) \to W) = (g' : C(V') \to W')$. □

Lemma 2.6. *Let X be a d-dimensional scheme over field k, $(g : C(V) \to W)$ a well defined family of d-dimensional cycles of X/k over W/k. Assume W is geometrically connected. Then $g^{[-1]}(w)$ is essentially independent of $w \in W$ i.e. for any $w, w' \in W$, we have*

$$g^{[-1]}(w) \overset{ess}{=} g^{[-1]}(w') \tag{31}$$

Remark 2.7. *We may assume W is irreducible.* ***The problem is that we should find at least a base field of this essentially same equation. It is natural to consider k to be the base field. While then it is hard to describe both of the special fiber and generic fiber.***

Theorem 2.1. *Let $(g : C(V) \to W)$ be a quasi-well defined family of cycles, W normal. Then $(g : C(V) \to W)$ is in fact well defined family.*

Proof. Since $(g : C(V) \to W)$ is a quasi-well defined family, we can define $\widetilde{V}$ and W' as proof of Proposition 2.5. Remains to check for the second essentially same condition. Set $Y := g^{-1}(w)$ and consider

$$(g' : \sum_i m_i[V_i' \times_{W'} p^{-1}(w)] \longrightarrow p^{-1}(w)) \tag{32}$$

Denote $V' = \cup_i V_i' \times_{W'} p^{-1}(w)$ with reduced scheme structure and $C(V') = \sum_i m_i[V_i' \times_{W'} p^{-1}(w)]$. Since g is proper and $\widetilde{g}_i$ is flat, g' is proper and g_i' is flat so that generic point of V' maps to generic point of $p^{-1}(w)$.

And for any $w' \in p^{-1}(w)$, the fiber of g_i' at w is $V_i' \times_{W'} w' \subset g^{-1}(w) \times_w w$, which is of dimension at most d. **If we can prove $(g' : C(V') \to p^{-1}(w))$ is a well defined family? It is really difficult to believe this is ture since our construction guarantees nothing about the dimension of fiber.** By Zariski's Main Theorem, we know $p^{-1}(w)$ is connected. Hence by Lemma 2.6, we see ${g'}^{[-1]}(w')$ is essentially independent of $w' \in p^{-1}(w)$. **Then try to show that the cycle-theoretic fiber here is just the cycle in the essentially same condition, which is another obstruction since V_i' are not irreducible and hence flatness of g_i' does not help us with cycle-theoretic fiber** □

2.3 Chow pullback

Let $(g : C(V) \to W)$ be a well defined family of cycles. Assume $f : W' \to W$ is a morphism with W' reduced, satisfying that $g^{[-1]}(f(w'))$ is defined over $k(f(w'))$ for all generic point $w' \in W'$. Then one can define a well defined family of cycles over W' as the following lemma.

Lemma 2.7. *Notations as above. Take V_j' to be irreducible components of $\widetilde{V} := V \times_W W'$ with reduced scheme structure. Set $V_{j,w'}'$ the fiber of $V_j' \to W'$ at generic point $w' \in W'$. Assume w' maps to w and $g^{[-1]}(w) = \sum_k a_k[Z_k]$, where Z_k vary over all d-dimensional irreducible closed subsets of $g^{-1}(w)$. Then*

- *Z_k are irreducible components of $g^{-1}(w)$ and $[Z_k \times_w w'] = \sum_j b_{k,j}[V_{j,w'}']$ for some integers $b_{k,j}$.*

Define $m'_j := \sum_{w' \ generic} \sum_k a_k b_{k,j}$. *Set* $f^{\mathrm{ch}}V := \cup_{j \ for \ m'_j \neq 0} V'_j$ *with reduced scheme structure,* $f^{\mathrm{ch}}g : f^{\mathrm{ch}}V \hookrightarrow \widetilde{V} \xrightarrow{\widetilde{g}} W'$ *and* $f^{\mathrm{ch}}C(V) = C(f^{\mathrm{ch}}V) := \sum_j m'_j[V'_j]$. *Then*

- $(f^{\mathrm{ch}}g : C(f^{\mathrm{ch}}V) \to W')$ *is a well defined family of* d*-dimensional cycles over* W'.

called the Chow pullback of $(g : C(V) \to W)$ *along* f. *We may also denote it by* $f^{\mathrm{ch}}(g : C(V) \to W)$.

Proof. By definition, $\dim g^{-1}(w) = d$ so that Z_k are irreducible components. Since w' is generic point, $V'_{j,w'}$ would be either empty or an irreducible component of $\widetilde{g}^{-1}(w')$. For those irreducible $V'_{j,w'}$, its image under $\widetilde{g}^{-1}(w') \to g^{-1}(w)$ is also irreducible. Note that $k(w') \to k(w)$ is flat of relative dimension 0 and g is proper, we see that those $V_{j,w'}$ of dimension d would map to some Z_k.

By some useful result, $Z_k \times_w w'$ is of pure dimension d. Thus $[Z_k \times_w w'] = \sum_j b_{k,j}[V'_{j,w'}]$ for some integers $b_{k,j}$. For those $V'_{j,w'}$ of dimension less than d, we assign $b_{k,j} = 0$ for all k. Thus $m'_j \neq 0$ if and only if there is some $V'_{j,w'}$ of dimension d, implying that $\dim(f^{\mathrm{ch}}g_j^{-1}(w')) = d$ and generic points of $f^{\mathrm{ch}}V$ all map to generic points of W'.

By upper semicontinuity, we get $\dim(f^{\mathrm{ch}}g_j^{-1}(w''))$ for all $w'' \in W'$. While $f^{\mathrm{ch}}g^{-1}(w'') \subseteq \widetilde{g}^{-1}(w'') = V \times_W w'' = g^{-1}(f(w'')) \times_{f(w'')} w''$ as sets, and $\dim(g^{-1}(f(w'')) \times_{f(w'')} w'') = d$, we get $\dim(f^{\mathrm{ch}}g_j^{-1}(w'')) \leq d$. Thus for all $w'' \in W'$, we get $\dim(f^{\mathrm{ch}}g_j^{-1}(w'')) = d$. As properness is stable under base change and composition, we get $f^{ch}g$ is proper. Hence $(f^{\mathrm{ch}}g : C(f^{\mathrm{ch}}V) \to W')$ is a quasi-well defined family of d-dimensional cycles.

Remains to show it is also a well defined family. For convenience, we forget part of notations above and denote V'_j to be irreducible components of $f^{\mathrm{ch}}V$. Same as construction in proof of Proposition 2.5, there is a commutative diagram

$$\begin{array}{ccc} V''_i & \longrightarrow & V \\ \downarrow{\scriptstyle g''_i} & & \downarrow{\scriptstyle g} \\ W'' & \xrightarrow{p} & W \end{array} \tag{33}$$

where $p : W'' \to W$ is a proper, surjective morphism and $V''_i \subset \widetilde{V} \times_W W''$ closed subschemes flat over W'' for all i, satisfying that

- for all $w \in W_0$ and $w'' \in p^{-1}(w)$, we have

$$\sum_i m_i[g_i''^{-1}(w'')] \overset{ess}{=} \sum_i m_i[g_i^{-1}(w)] \tag{34}$$

- for all $w \in W$ and $w''_1, w''_2 \in p^{-1}(w)$, we have

$$\sum_i m_i[g_i''^{-1}(w''_1)] \overset{ess}{=} \sum_i m_i[g_i''^{-1}(w''_2)] \tag{35}$$

Pullback along $f : W' \to W$, we get commutative diagram

$$\begin{array}{ccc} V''_i \times_W W' & \longrightarrow & \widetilde{V} \times_W W' \\ \downarrow{\scriptstyle \widetilde{g}_i} & & \downarrow{\scriptstyle \widetilde{g}} \\ W'' \times_W W' & \xrightarrow{p'} & W' \end{array} \tag{36}$$

where $\widetilde{V}\times_W W'$ is proper over W', $p': W''\times_W W' \to W'$ is a proper, surjective morphism and $V_i''\times_W W' \subset \widetilde{V}\times_W W''\times_W W'$ closed subschemes flat over W'' for all i. Remains to prove the two essentially same conditions. For all generic point $w'\in W'$ and $w'''\in p'^{-1}(w')$, assume w' maps to w and w''' maps to $w''\in p^{-1}(w)$. By Proposition 2.5, cycle over w'' computes $g^{[-1]}(w)$ i.e.

$$\sum_i m_i[g_i''^{-1}(w'')] \overset{ess}{=} g^{[-1]}(w) \tag{37}$$

so that

$$\sum_i m_i[g_i''^{-1}(w'')\times_{w''} w'''] \overset{ess}{=} g^{[-1]}(w)\times_w w''' \overset{ess}{=} (g^{[-1]}(w)\times_w w')\times_{w'} w''' \text{ over } k(w) \tag{38}$$

By Corollary 2.4, they are also essentially same over $k(w')$. Note that $g_i''^{-1}(w'')\times_{w''} w''' \cong \widetilde{g}_i^{-1}(w''')$, we get

$$\begin{aligned}\sum_i m_i[g_i''^{-1}(w''')] &\overset{ess}{=} g^{[-1]}(w)\times_w w' \\ &= \sum_{j, V'_{j,w'}\neq\varnothing} m_j'[g_i^{-1}(w)\times_w w'] \\ &= \sum_j m_j'[g_i^{-1}(w)\times_w w']\end{aligned} \tag{39}$$

the first essentially condition holds. For the second one, note that both sides of the equality are pullback from fiber of g_i'', it is clear that they are still essentially same after base change, done! □

Remark 2.8. *During the proof, many components would be abandoned for example those w' not mapping to generic point of W or those $V'_{j,w'}$ of dimension less than d. This is compatible with our original idea of a well defined family of cycles.*

In addition, though we pick the diagrams 33 in the proof, in fact for any family of diagrams satisfying the conditions in Proposition 2.5 associated to $(g: C(V)\to W)$, after base change, the new family of diagrams is associated to $(f^{\mathrm{ch}}g: f^{\mathrm{ch}}C(V)\to W')$.

Lemma 2.8. *Let $(g: C(V)\to W)$ be a well defined family, $f_1: W_1\to W$ and $f_2: W_2\to W_1$ morphisms with W_1 and W_w reduced. Then*

$$f_2^{\mathrm{ch}}\circ f_1^{\mathrm{ch}}(g: C(V)\to W) = (f_1\circ f_2)^{\mathrm{ch}}(g: C(V)\to W) \tag{40}$$

Proof. Since $(g: C(V)\to W)$ is a well defined family, there are commutative diagrams

$$\begin{array}{ccc} V_j' & \longrightarrow & \widetilde{V} \\ \big\downarrow{\scriptstyle \widetilde{g}_j} & & \big\downarrow{\scriptstyle \widetilde{g}} \\ W' & \xrightarrow{p} & W \end{array} \tag{41}$$

where $\widetilde{V}$ is a scheme proper over W such that g factors through $\widetilde{g}$ by a closed immersion $V\hookrightarrow\widetilde{V}$, $p: W'\to W$ is a proper, surjective morphism and $V_j'\subset\widetilde{V}\times_W W'$ closed subschemes flat over W' for all j, and there are integers m_j' such that

- for all w_g generic and $w' \in p^{-1}(w_g)$, we have

$$\sum_j m'_j[\widetilde{g}_j^{-1}(w')] \overset{ess}{=} \sum_i m_i[g_i^{-1}(w_g)] \tag{42}$$

- for all $w \in W$ and $w'_1, w'_2 \in p^{-1}(w)$, we have

$$\sum_j m'_j[\widetilde{g}_j^{-1}(w'_1)] \overset{ess}{=} \sum_j m'_j[\widetilde{g}_j^{-1}(w'_2)] \tag{43}$$

After base change along f_1, we get diagrams

$$\begin{array}{ccc} V'_j \times_W W_1 & \longrightarrow & \widetilde{V} \times_W W_1 \\ \downarrow{\scriptstyle \widetilde{g}^1_j} & & \downarrow{\scriptstyle \widetilde{g}^1} \\ W' \times_W W_1 & \xrightarrow{p^1} & W_1 \end{array} \tag{44}$$

and again after base change along f_2, we get

$$\begin{array}{ccc} V'_j \times_W W_2 & \longrightarrow & \widetilde{V} \times_W W_2 \\ \downarrow{\scriptstyle \widetilde{g}^2_j} & & \downarrow{\scriptstyle \widetilde{g}^2} \\ W' \times_W W_2 & \xrightarrow{p^2} & W_2 \end{array} \tag{45}$$

Note that diagrams 44 are associated to $(f_1^{\mathrm{ch}} g : f_1^{\mathrm{ch}} C(V) \to W_1)$, diagrams 45 are associated both to $(f_1 \circ f_2)^{\mathrm{ch}}(g : C(V) \to W)$ and $f_2^{\mathrm{ch}} \circ f_1^{\mathrm{ch}}(g : C(V) \to W)$. By Lemma 2.5, we are done! $\square$

3 Seminormal and Weakly Normal

Definition 3.1. *Let $f : X \to Y$ be a finite surjective morphism between reduced schemes. Define the seminormalization (resp. weak normalization) of Y with respect to f to be a scheme $\widetilde{Y}_f^{\mathrm{semi}}$ (resp. $\widetilde{Y}_f^{\mathrm{weak}}$) together with a finite bijective morphism $\widetilde{f} : \widetilde{Y} \to Y$ such that*

- *f factors through $\widetilde{f}$*
- *for all $y' \in \widetilde{Y}_f^{\mathrm{semi}}$, $k(\widetilde{f}(y')) \to k(y')$ is isomorphic.*
- *(resp. for all $y' \in \widetilde{Y}_f^{\mathrm{weak}}$, $k(\widetilde{f}(y')) \to k(y')$ is purely inseparable extension and if $\widetilde{f}(y)$ is generic point, then isomorphic)*
- *if $f' : Y' \to Y$ satisfies properties above, then $\widetilde{f}$ uniquely factors through f'*

For excellent reduced Y, the normalization map $\widetilde{Y} \to Y$ is finite surjection. Then we define the seminormalization (resp. weak normalization) of Y to be the seminormalization (resp. weak normalization) of Y with respect to $\widetilde{Y} \to Y$, denoted by $\widetilde{Y}^{\mathrm{semi}}$ (resp. $\widetilde{Y}^{\mathrm{weak}}$). If $Y \cong \widetilde{Y}^{\mathrm{semi}}$ (resp. $Y \cong \widetilde{Y}^{\mathrm{weak}}$), then we say Y is seminormal (resp. weakly normal).

Remark 3.1. *Clearly, seminormal implies weakly normal and when characteristic 0, seminormalization is same to weak normalization.*

From now on, schemes would be assumed to be excellent.

Lemma 3.1. *Let $f : X \to Y$ be a finite surjective morphism between reduced schemes. Then $\widetilde{Y}_f^{\text{semi}}$ and $\widetilde{Y}_f^{\text{weak}}$ are both reduced.*

Proof. Here we only proves for seminormalization, same argument can be applied to weak normalization. Since the reduction map is a bijective closed immersion hence finite surjective morphism, by universal property of reduction, we get $\operatorname{red} \widetilde{Y}_f^{\text{semi}} \to Y$ also satisfies the conditions. Hence by universal property of seminormalization, $\widetilde{Y}_f^{\text{semi}} \to Y$ uniquely factors through $\operatorname{red} \widetilde{Y}_f^{\text{semi}} \to Y$. Thus there is a commutative diagram

$$\begin{tikzcd} \operatorname{red} \widetilde{Y}_f^{\text{semi}} \arrow[rr, shift left, "i"] \arrow[dr] & & \widetilde{Y}_f^{\text{semi}} \arrow[ll, shift left, "j"] \arrow[dl] \\ & Y & \end{tikzcd} \tag{46}$$

Then by universal property of reduction and seminormalization, we get $j \circ i = \text{id}$ and $i \circ j = \text{id}$. Conclude that $\widetilde{Y}_f^{\text{semi}} \cong \operatorname{red} \widetilde{Y}_f^{\text{semi}}$ and hence reduced. □

Lemma 3.2. *Let $f : Y \to X$ be a morphism between reduced schemes, Y seminormal. Then f uniquely factors as $Y \to \widetilde{X}^{\text{semi}} \to X$.*

Proof. Consider the fibered diagram

$$\begin{tikzcd} \widetilde{X}^{\text{semi}} \times_X Y \arrow[r, "g'"] \arrow[d, "f'"] & Y \arrow[d, "f"] \\ \widetilde{X}^{\text{semi}} \arrow[r, "g"] & X \end{tikzcd} \tag{47}$$

Since g is finite surjective, we get g' is also finite surjective. For $y \in Y$, assume y maps to x and $g^{-1}(x) = x'$. Consider the fibered diagram

$$\begin{tikzcd} Y \times_X x' \arrow[r, "g'_x"] \arrow[d, "f'_x"] & Y \times_X x \arrow[d, "f_x"] \\ x' = \widetilde{X}^{\text{semi}} \times_X x \arrow[r, "g_x"] & x \end{tikzcd} \tag{48}$$

By definition, g_x is isomorphic and hence f'_x is isomorphic so that g' is bijective. Denote $y' := g'^{-1}(y)$. Then the underlying topological space of $x' \times_x y = \widetilde{X}^{\text{semi}} \times_X y$ is just $\{y'\}$. As $x' \times_x y = \operatorname{Spec}(k(x') \otimes_{k(x)}) k(y) \cong \operatorname{Spec} k(y)$, we get $k(y') \cong k(y)$. In particular, for any generic point $y_g \in Y$ and $y'_g = g'^{-1}(y_g)$, $k(y'_g) \cong k(y_g)$ so that by some useful result, the normalization map of Y factors through g'. Then by universal property of seminormalization, we see g' is isomorphic and hence f factors as $Y \to \widetilde{X}^{\text{semi}} \to X$. □

Let $X \to S$ be a smooth and projective morphism of relative dimension N, $(g : C(V) \to W)$ well defined family of $(N-1)$-dimensional cycles of X/S over W/S. Assume w_g is a generic point of W. Since $X \times_S W \times_W k(w_g)^{p^{-\infty}}$ is smooth, by Proposition 2.1, $g^{[-1]}(w_g)$ is fundamental cycle of some Cartier divisor $D(w_g)$ on $X \times_S W \times_W k(w_g)^{p^{-\infty}}$. As $D(w_g)$ is flat over $k(w_g)^{p^{-\infty}}$, it corresponds to a morphism $\operatorname{Spec} k(w_g) \to \operatorname{CDiv}(X \times_S W/W)$. Take the schematic image W' of morphism $k(w_g) \to \mathrm{CDiv}(X \times_S W/W)$. And $p : W' \to W$ the projection.

Lemma 3.3. *Notations as above, we have that*

- *W' is dominated by the weak normalization of W.*
- *Assume W' is seminormal. Then p is isomorphic if and only if $g^{[-1]}(w_g)$ is defined over $k(w_g)$.*
- *If W is weakly normal, then g is flat.*

4 Chow Varieties

Now it's time for us to define Chow functor and prove its representability when characteristic 0.

Definition 4.1. *Let X be a scheme over S. For all reduced scheme W over S, define*

$$\mathcal{C}how(X/S)(W) := \{\text{well defined family of effective cycles of } X/S \text{ over } W/S\}$$

If X moreover is projective over S with line bundle $\mathcal{O}_X(1)$, define

$$\mathcal{C}how_{d,d'}(X/S)(W) := \{(g : C(V) \to W) \in \mathcal{C}how(X/S)(W) \text{ of dimension } d \text{ and degree } d'\}$$

Hence by Proposition 2.4, for connected W, we have

$$\mathcal{C}how(X/S)(W) = \sqcup_{d,d'} \mathcal{C}how_{d,d'}(X/S)(W)$$

Theorem 4.1. *Notations as above. There is contravariant functor*

$$\mathcal{C}how_{d,d'}(X/S)(\cdot) : \{\text{seminormal } S\text{-schemes}\} \longrightarrow \mathbf{Set} \tag{49}$$

And the functor is represented by a seminormal scheme $\mathrm{Chow}_{d,d'}(X/S)$ which is projective over S, with universal family $\mathrm{Univ}_{d,d'}(X/S)$.

Functoriality immediately comes from the results in Chow pullback. We will prove the representability step by step. To be clarified, during the proof, we would not assume characteristic 0 so that it is clearer to see the role of characteristic 0.

Step 1: Construction of Chow map

In this step, for $\mathbb{P}_S$ the projective bundle over S associated to some rank $n+1$ vector bundle, we construct a natural transfromation from $\mathcal{C}how_{d,d'}(\mathbb{P}_S/S)$ to a Chow functor parameterizing hypersurfaces.

Let $\mathbb{P}_S^\vee$ be the dual bundle. Consider the incidence correspondence

$$I := \{(x, H_1, \cdots, H_{d+1}) \in \mathbb{P}_S \times_S (\mathbb{P}_S^\vee)^{d+1} \big| x \in H_1 \cap \cdots \cap H_{d+1}\} \tag{50}$$

Denote $p_1 : I \to \mathbb{P}_S$ and $p_2 : I \to (\mathbb{P}_S^\vee)^{d+1}$ the projections. Notice that p_1 and p_2 are both smooth morphisms.

Given $s : W \to S$ and well defined family $(g : C(V) \to W)$ of $\mathbb{P}_S/S$ over W/S, there is a diagram

$$\begin{tikzcd} V \arrow[r, hook] & s^*\mathbb{P}_S & s^*I \arrow[l, "s^*p_1"'] \arrow[d, "s^*p_2"] \\ & & s^*(\mathbb{P}_S^\vee)^{d+1} \end{tikzcd} \tag{51}$$

For quasi-cycle Z on V, set $\mathrm{Ch}(Z) := (s^*p_2)_* \circ (s^*p_1)^*(Z)$. Also define $\mathrm{Ch}(V_i) = \mathrm{red}(s^*p_2)_* \circ (s^*p_1)^*(V_i)$.

Lemma 4.1. *The projection $p : (s^*p_1)^*V_i \to \mathrm{Ch}(V_i)$ is generically finite and generically injective i.e. for an open dense subset U of $\mathrm{Ch}(V_1)$, the morphism $p^{-1}(U) \to U$ is finite and injective. In particular, when characteristic 0, we get $\mathrm{Ch}([V_i]) = [\mathrm{Ch}(V_i)]$.*

Lemma 4.2. *For $W = \mathrm{Spec}\, k$ spectrum of a field, the projection $p : (s^*p_1)^*V_i \to \mathrm{Ch}(V_i)$ is a local isomorphism at $(x, H_1, \cdots, H_{d+1})$ if and only if*

$$V_i \cap H_1 \cap \cdots \cap H_{d+1} = \{x\} \tag{52}$$

as schematic intersection.

Proposition 4.1. *Notations as above, then we have*

- *$(\mathrm{Ch}(g) : \mathrm{Ch}(C(V)) \to W)$ is a well defined family of $(dn + n - 1)$-dimensional cycles of $(\mathbb{P}_S^\vee)^{d+1}/S$ over W/S, for convenience, we may this is a well defined family of hypersurfaces.*
- *$\mathrm{Ch}(C(V)) = \sum_i m_i k_i[\mathrm{Ch}(V_i)]$ for some integers k_i. If every geometric generic fiber of g is reduced, then $k_i = 1$ for all i. In particular, this holds if characteristic 0.*
- *Let $f : W' \to W$ be a morphism between reduced schemes. Then*

$$(\mathrm{Ch}(f^{\mathrm{ch}}g) : \mathrm{Ch}(f^{\mathrm{ch}}C(V)) \to W') \cong (f^{\mathrm{ch}}\,\mathrm{Ch}(g) : f^{\mathrm{ch}}\,\mathrm{Ch}(C(V)) \to W') \tag{53}$$

- *There is a unique closed subscheme $\mathrm{Ch}(V) \subset (\mathbb{P}_S^\vee)^{d+1}$ such that $\mathrm{Ch}(V)$ is purely one-codimensional, has no embedded point and its fundamental cycle is $\mathrm{Ch}(C(V))$.*

Step 2: Chow forms

Definition 4.2. *Let K/k be a field extension, $Z \in Z_d(\mathbb{P}_K)$. Then $\mathrm{Ch}(Z)$ is hypersurface of $(\mathbb{P}_K^\vee)^{d+1}$ defined by a multihomogeneous polynomial $\mathrm{ch}(Z)$, called the Chow form or Cayley form of cycle Z. In addition, the minimal field of definition $\mathrm{Ch}(Z)$ over k is called the chow field of Z, denoted by $k^{\mathrm{ch}}(Z)$.*

Lemma 4.3. *Let $Z_= \sum_i a_i[V_i] \in Z_d(\mathbb{P}_K)$. Then*

- *If $E \supset K$ is a field extension, then $\mathrm{ch}(Z_E) = \mathrm{ch}(Z_K)$.*

- *There is a one-to-one correspondence*

$$\{components\ of\ Z\} \leftrightsquigarrow \{irreducible\ divisor\ of\ \mathrm{ch}(Z)\} \tag{54}$$

  In fact, we have $\mathrm{ch}(Z) = \prod_i (\mathrm{ch}([V_i]))^{a_i}$.

- *Set* $d' = \deg_{\mathcal{O}(1)}$, *then* $\mathrm{ch}(Z)$ *is of multidegree* $(d', \cdots, d')$.

Corollary 4.1. *Assume* S *is spectrum of a field of characteristic* 0. *Denote* $q : (\mathbb{P}_S^\vee)^{d+1}$ *the projection and* $\mathbb{H}_S := \mathrm{Proj}_S(q_* \mathcal{O}_{(\mathbb{P}_S^\vee)^{d+1}}(d', d', \cdots, d'))$, *which parameterizes hypersurfaces of multidegree* $(d', d', \cdots, d')$ *in* $(\mathbb{P}_S^\vee)^{d+1}$. *Then there is a natural transformation*

$$\mathrm{Ch} : \mathcal{C}how_{d,d'}(\mathbb{P}_S/S)(\cdot) \longrightarrow \mathrm{Hom}(\cdot, \mathbb{H}) \tag{55}$$

Proof. By Proposition 4.1, we sends $(g : C(V) \to W)$ to a hypersurface $\mathrm{Ch}(V)$, which by Lemma 3.3 is flat. Hence it corresponds to a morphism $W \to \mathrm{Hilb}((\mathbb{P}_S^\vee)^{d+1}/S)$. And by Lemma 4.3, we know the chow form $\mathrm{ch}(C(V))$ is of multidegree $(d', \cdots, d')$. Hence the morphism factors through $\mathbb{H}$. □

We also define inverse map of Ch.

Definition 4.3. *First set-theoretically define*

$$\mathrm{Ch}^{-1}(H) := \{x \in \mathbb{P}_S \big| p_1^{-1}(x) \subset p_2^{-1}(H)\} \tag{56}$$

for hypersurface $H \subset (\mathbb{P}_S^\vee)^{d+1}$, *which is a closed subset of* $\mathbb{P}_S$ *since* p_1 *is flat locally of finite presentation and hence an open map.*

Proposition 4.2. *For all hypersurface* $H \subset (\mathbb{P}_S^\vee)^{d+1}$, $\dim(\mathrm{Ch}^{-1}(H)) \leq d$. *And for irreducible and reduced* H, *if* $\dim(\mathrm{Ch}^{-1}(H)) = d$, *then* $\mathrm{Ch}^{-1}(H)$ *is irreducible of dimension* d. *And if moreover* $S = \mathrm{Spec}\, k$ *for some perfect field* k, *then* $\mathrm{Ch}([\mathrm{Ch}^{-1}(H)]) = H$, *where* $[\mathrm{Ch}^{-1}(H)]$ *is fundamental cycle of* $\mathrm{Ch}^{-1}(H)$ *with the reduced scheme structure.*

Corollary 4.2. *Assume* k *is perfect, then there is a one-to-one correspondence*

$$\{d\text{-}dimensional\ cycles\ of\ degree\ d'\ in\ \mathbb{P}_k\} \leftrightsquigarrow \left\{\begin{matrix} H \in \mathbb{H}_k\ with\ \dim(\mathrm{Ch}^{-1}(H_j)) = d \\ for\ all\ irreducible\ components\ H_j\ of\ H \end{matrix}\right\}$$

In particular, this holds when characteristic 0.

Proof. For H as above, assume $[H] = \sum_j a_j H_j$ and set $\mathrm{Ch}^{-1}([H]) = \sum_j a_j [\mathrm{Ch}^{-1}(H_j)]$. Check that Ch^{-1} is then inverse of *chowmap*. □

Step 3: Construction of universal family

As a preliminary step, set

$$\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S) := \{H \in \mathbb{H}_S \big| \dim(\mathrm{Ch}^{-1}(H_j)) = d, \forall \text{ irreducible components } H_j \text{ of } H\}$$

Let $h : \mathbb{H}_S \to S$ be the projection map. Take universal family $\mathbb{U}_S$ of $\mathbb{H}_S$ which is a hypersurface of $(h^*\mathbb{P}_S^\vee)^{d+1}$. There is a diagram

$$\begin{tikzcd} h^*\mathbb{P}_S & h^*I \arrow[l, "h^*p_1"'] \arrow[d, "h^*p_2"] \\ & (h^*\mathbb{P}_S^\vee)^{d+1} \end{tikzcd} \tag{57}$$

By Proposition 4.2, we know $\dim(\mathrm{Ch}^{-1}(\mathbb{U}_S)) \leq d$. And there is a projection map $\pi : \mathrm{Ch}^{-1}(\mathbb{U}_S) \to \mathbb{H}_S$. Hence each topological fiber of π has dimension at most d.

Lemma 4.4. *We have an equivalent description of* $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$ *that*

$$\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S) = \{h \in \mathbb{H}_S \big| \pi^{-1}(H) \textit{ has pure dimension } d\} \tag{58}$$

Proposition 4.3. $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$ *is a closed subset of* $\mathbb{H}_S$*.*

When S is spectrum of a field, Corollary 4.2 tells us that the morphism $W \to \mathbb{H}$ corresponding to $(g : C(V) \to W)$, given by Corollary 4.1, would again factor through $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$.

Now for general S, let W be a seminal scheme and $s : W \to S$ a morphism. Assume $(\widetilde{h} : C(\mathbf{H}) \to W)$ is a well defined family of hypersurfaces of $\mathbb{P}_S/S$ over W/S such that for any field F with morphism $\mathrm{Spec}\, F \to W$, the determining polynomial of hypersurface $\mathbf{H} \times_W F$ is Chow form of some cycle in $Z_d(\mathbb{P}_F)$.

In order to define $\mathrm{Ch}^{-1}(C(\mathbf{H}))$, we need to assign multiplicities of irreducible components.

Lemma 4.5. *Define* $\mathrm{Ch}^{-1}(C(\mathbf{H}))$ *as following. Then* $\mathrm{Ch}^{-1}(\widetilde{h} : C(\mathbf{H}) \to W) := (\mathrm{Ch}^{-1}(\widetilde{h}) : \mathrm{Ch}^{-1}(C(\mathbf{H})) \to W)$ *is a well defined family of cycles. In addition,* Ch^{-1} *is the inverse of Chow map* Ch*.*

Proof. To do this, we may localize at generic point of $w_g \in W$. By assumption, there is a unique cycle over $\mathbb{P}_{k(w_g)}$ with Chow form determining $\mathbf{H} \times_W k(w_g)$. This would assign the multiplicity. □

Proposition 4.4. *When characteristic* 0*, for* $f : W' \to W$ *morphism between seminormal schemes, we have that*

$$\mathrm{Ch}^{-1} \circ f^{\mathrm{ch}}(\widetilde{h} : C(\mathbf{H}) \to W) \cong f^{\mathrm{ch}} \circ \mathrm{Ch}^{-1}(\widetilde{h} : C(\mathbf{H}) \to W) \tag{59}$$

Definition 4.4. *Take* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$ *to be the seminormalization of* $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$*. In particular,* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$ *is seminormal and projective over* S*. Set*

$$\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S) := \mathrm{Ch}^{-1}(\mathbb{U}_S \times_{\mathbb{H}_S} \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$$

and

$$C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) := \mathrm{Ch}^{-1}(C(\mathbb{U}_S \times_{\mathbb{H}_S} \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)))$$

which together give a well defined family $(u : C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$*, called universal well defined family of* $\mathbb{P}_S/S$*.*

Remark 4.1. *By Lemma 3.2, we know that a morphism* $W \to \mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$ *from a seminormal scheme* W *would factor through* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$*. Hence we finally get a natural transformation from* $\mathcal{C}how_{d,d'}(\mathbb{P}_S/S)(\cdot)$ *to* $\mathrm{Hom}(\cdot, \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$*.*

Now for X projective over S, choose an embedding $X \hookrightarrow \mathbb{P}_S$. Let

$$\mathrm{Chow}'_{d,d'}(X/S) := \{z \in \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S) \big| u^{-1}(z) \subset X \times_S k(z)\}$$

Lemma 4.6. $\mathrm{Chow}'_{d,d'}(X/S)$ *is a closed subset of* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$*.*

Take $\mathrm{Chow}_{d,d'}(X/S)$ to be the seminormalization of $\mathrm{Chow}'_{d,d'}(X/S)$. Define the universal well defined family $(u_X : C(\mathrm{Univ}_{d,d'}(X/S)) \to \mathrm{Chow}_{d,d'}(X/S))$ of X/S to be the Chow pullback of well defined family of $\mathbb{P}_S/S$ along $\mathrm{Chow}_{d,d'}(X/S) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$. Clearly, $\mathrm{Chow}_{d,d'}(X/S)$ is seminormal and projective over S.

Step 4: Complete the proof

Remains to use the universal family to show the representability when characteristic 0. First check that $\mathcal{C}how_{d,d'}(\mathbb{P}_S/S)$ is representable by $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$. As we remarked above, there is a natural transformation

$$\varphi : \mathcal{C}how_{d,d'}(\mathbb{P}_S/S)(\cdot) \longrightarrow \mathrm{Hom}(\cdot, \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)) \tag{60}$$

Also for any seminormal W over S, by Lemma 4.5, $\varphi(W)$ is a bijection. It is clear that $(u : C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$ is the universal family.

Now for X projective over S, choose embedding $X \hookrightarrow \mathbb{P}_S$. As $X \times_S W$ is a closed subscheme of $\mathbb{P}_S \times_S W$, for all well defined family $(g : C(V) \to W)$ of X/S over W/S, it is also a well defined family of $\mathbb{P}_S/S$ over W/S. Hence it corresponds to a morphism $f : W \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$ and

$$(g : C(V) \to W) = f^{\mathrm{ch}}(u : C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)) \tag{61}$$

Then by construction of Chow pullback, we know for all generic point $w_g \in W$, either $u^{-1}(f(w_g)) = \varnothing$ or $u^{-1}(f(w_g)) \times_{f(w_g)} w_g = V \times_W w_g \subset X \times_S w_g$. While $u^{-1}(f(w_g)) \times_{f(w_g)} w_g$ surjectively maps to $u^{-1}(f(w_g))$, we get $f(w_g) \in \mathrm{Chow}'_{d,d'}(X/S)$. Thus f factors through

$\mathrm{Chow}'_{d,d'}(X/S)$. Again by Lemma 3.2, f factors through $\mathrm{Chow}_{d,d'}(X/S)$. Finally we get a natural transformation

$$\varphi' : \mathcal{C}how_{d,d'}(X/S)(\cdot) \longrightarrow \mathrm{Hom}(\cdot, \mathrm{Chow}_{d,d'}(X/S)) \tag{62}$$

Note that $\varphi'(W)$ is restriction of the map $\varphi(W)$ hence injective. For surjectivity, since each morphism $W \to \mathrm{Chow}_{d,d'}(X/S)$ also gives a morphism in $\mathrm{Hom}(W, \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$. Hence there is a well defined family $(g : C(V) \to W)$ of $\mathbb{P}_S/S$ over W/S, which is the Chow pullback of universal well defined family of $\mathbb{P}_S/S$ along $W \to \mathrm{Chow}_{d,d'}(X/S) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$.

We may first pull back along $\mathrm{Chow}'_{d,d'}(X/S) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$. Denote the Chow pullback $(g' : C(V') \to \mathrm{Chow}'_{d,d'}(X/S))$. Then $V \subseteq V' \times_{\mathrm{Chow}'_{d,d'}(X/S)} W$. While by definition, $V' \subseteq X \times_S \mathrm{Chow}'_{d,d'}(X/S)$, we get $V \subseteq X \times_S W$. Thus $(g : C(V) \to W)$ is in fact a well defined family of X/S over W/S. Conclude that $\varphi'(W)$ is bijective so that $\mathcal{C}how_{d,d'}(X/S)(\cdot)$ is representable by $\mathrm{Chow}_{d,d'}(X/S)$. As $\mathrm{Univ}_{d,d'}(X/S)$ is the Chow pullback of $\mathrm{Univ}_{d,d'}(\mathbb{P}_S \times_S S)$, it is the universal family of φ'.

Obstructions in positive characteristic

After proving the representability when characteristic 0, let's find out where the obstructions in positive characteristic exist. When the characteristic of the base field is $p > 0$, the elegant correspondence between algebraic cycles and their Chow forms given by Corollary 4.2 breaks down, so we cannot directly get a bijection, introducing severe obstructions to representability.

- **Purely Inseparable Extensions:** The primary obstruction arises from purely inseparable field extensions. If an algebraic cycle is covered by a family where the generic fiber is geometrically non-reduced (which can happen in characteristic p due to purely inseparable morphisms), the classical Chow form fails to accurately capture the multiplicity of the cycle.

- **Topological vs. Scheme-Theoretic Fibers:** Because the Chow form can evaluate to a p^e-th power for inseparable families, the classical definition only determines the cycle up to a Frobenius twist. The natural Hilbert-Chow morphism Hilb $\to$ Chow may have purely inseparable fibers, meaning the Chow variety, as classically constructed, is functionally closer to a topological space than a space carrying the correct scheme-theoretic moduli structure.

To resolve these obstructions in modern algebraic geometry, the functorial definition of families of cycles must be modified. Modern proofs (such as those by Rydh [2]) utilize **divided powers** (or operations in the ring of Witt vectors) to keep track of multiplicities correctly, allowing for a well-behaved moduli stack or scheme even in mixed and positive characteristics.

In Kollár's book, he also defines several possible generalizations of Chow functor.

Definition 4.5. *Let $(g : C(V) \to W)$ be a well defined family of cycles.*

- *We say it satisfies the Chow field condition if*

$$k^{\mathrm{ch}}(g^{[-1]}(w)) = k(w), \forall w \in W \tag{63}$$

- *We say it satisfies the field of definition condition if*

$$g^{[-1]}(w) \text{ defined over } k(w), \forall w \in W \tag{64}$$

Definition 4.6. *Let X be a scheme over S. With two conditions above, for reduced W, we define*

- $\mathcal{C}how^{\mathrm{big}}(X/S)(W) := \{$*well defined family of cycles of X/S over W/S*$\}$
- $\mathcal{C}how(X/S)(W) := \begin{Bmatrix} \text{well defined family of cycles of } X/S \text{ over } W/S \\ \text{satisfying the Chow field condition} \end{Bmatrix}$
- $\mathcal{C}how^{\mathrm{small}}(X/S)(W) := \begin{Bmatrix} \text{well defined family of cycles of } X/S \text{ over } W/S \\ \text{satisfying the field of definition condition} \end{Bmatrix}$

We can also define those subset fixing dimension and degree when X is projective over S.

Remark 4.2. *Clearly, these three definitions coincide when characteristic* 0.

Theorem 4.2. *Let X be a scheme projective over S with relative ample line bundle $\mathcal{O}(1)$. Then*

- $\mathcal{C}how_{d,d'}(X/S)(\cdot)$ *and* $\mathcal{C}how^{\mathrm{small}}_{d,d'}(X/S)$ *become functor under Chow pullback. And they are both coarsely representable by* $\mathrm{Chow}_{d,d'}(X/S)$.
- $\mathrm{Chow}_{d,d'}(X/S)$ *is semistable and projective over S.*
- $\mathrm{Chow}(X/S)$ *is independent of the choice of $\mathcal{O}(1)$.*

References

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[2] David Rydh, *Families of cycles and the Chow scheme*, PhD dissertation, KTH, Stockholm, 2008.